

Chef Classification from Recipe Text using DistilBERT

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October 6, 2025

1 Problem Setting

We tackle the task of assigning each recipe in a six-class corpus to its original chef using only textual information. The raw dataset contains 2,999 labelled recipes; 14 are duplicated verbatim. Removing those duplicates prior to splitting leaves 2,985 unique examples whose class counts range from 365 to 801. We generate an 80/20 stratified train/validation split and evaluate models with accuracy (for comparability with provided baselines) and macro-F1 (to monitor minority chefs). Our final DistilBERT pipeline achieves 92.13% validation accuracy and 0.916 macro-F1, comfortably beating the strongest TF-IDF baseline (43% accuracy).

2 Data Preparation

Field parsing and ordering. Ingredients, tags, and steps are parsed from their string representations into lists. We concatenate the five textual sections in the order *name* $\rightarrow$ *ingredients* $\rightarrow$ *tags* $\rightarrow$ *description* $\rightarrow$ *steps*. This decision protects identifying fields from truncation: after deduplication, steps account for about 46% of tokens, while the median sequence length is 234 tokens and the 95th percentile is 418, so most recipes fit within DistilBERT’s 512-token limit (Appendix Figure 3).

Leakage prevention. Deduplicating before splitting is essential. Early pilots performed the split first and overestimated accuracy by several points because identical recipes appeared in both splits. With leakage removed, class priors remain close to the original distribution and no recipe text is shared across folds.

Minimal normalisation. We deliberately avoid aggressive cleaning. DistilBERT’s uncased tokenizer already handles casing, while keeping punctuation and rare words preserves stylistic cues such as “diabetic cooking” or “OAMC freezer meal” that are strongly chef-specific.

3 Model and Optimisation Choices

Architecture. We fine-tune DistilBERT-base-uncased. The classification head is the default: dropout 0.2, a $768 \rightarrow 768$ projection with ReLU, and a linear layer producing six logits. A diagram of the full pipeline—from field concatenation to prediction—is provided in Appendix Figure 1. We experimented with alternative activations (GELU) and deeper heads, but these offered no consistent improvement once we tightened evaluation cadence.

Optimiser and schedule. Training uses

AdamW with learning rate 2×10^{-5} , weight decay 0.01, gradient clipping at 1.0, and a linear warm-up covering 6% of steps. To keep Mac hardware responsive we adopt “chill mode” batch sizes of 8 for training and 16 for evaluation with dynamic padding. We evaluate every 100 optimisation steps and apply early stopping after two non-improving macro-F1 evaluations, keeping the checkpoint with the highest macro-F1.

Monitoring. Narrowing the evaluation cadence was crucial. Fixed epoch counts suggested continued improvement, but step-wise logging reveals that accuracy and macro-F1 plateau around step 1300 (Appendix Figure 2). This insight guided the final schedule and avoided unnecessary epochs.

4 Results and Analysis

Table 1: Validation metrics compared with TF-IDF baselines.

Method	Accuracy	Macro-F1
TF-IDF (description)	30.0%	—
TF-IDF (all fields)	43.0%	—
DistilBERT (ours)	92.13%	0.916

Class-level precision and recall all exceed 0.85. Chef 1533 is slightly over-predicted and Chef 3288 slightly under-predicted, reflecting their shared focus on quick appetisers versus make-ahead dishes. Appendix Figure 4 shows that prediction proportions stay within ± 2.3 percentage points of the training priors. Remaining errors fall into three buckets: minimal context (very short recipes), cross-cultural dishes appearing in multiple cookbooks, and the small fraction of recipes truncated beyond 512 tokens.

Compared to TF-IDF, DistilBERT benefits from contextual semantics: phrases like “diabetic cooking” or ingredient substitutions combine with instruction style to distinguish chefs. The baseline gap is illustrated in Appendix Figure 5.

5 Design Rationale and Future Work

The final pipeline emphasises three principles. **(1) Protect high-signal fields:** ordering the concatenation keeps names and ingredients intact while long step lists absorb truncation. **(2) Control leakage and cadence:** deduplication plus step-wise early stopping produce reliable metrics and surfaced the optimal checkpoint near epoch 5. **(3) Prefer simplicity unless justified:** the default ReLU head and modest batch sizes already attain 92% accuracy; heavier heads or models offered no gain under the resource envelope.

Future extensions could integrate structured cues (ingredient counts, prep times), run cross-validation for tighter variance estimates, and perform attribution analyses to verify the model attends to stylistic signals rather than only obvious keywords. Exploring data augmentation (e.g., recipe paraphrasing) may further improve robustness on chefs with fewer samples.

Bibliography

References

Appendix: Supporting Figures

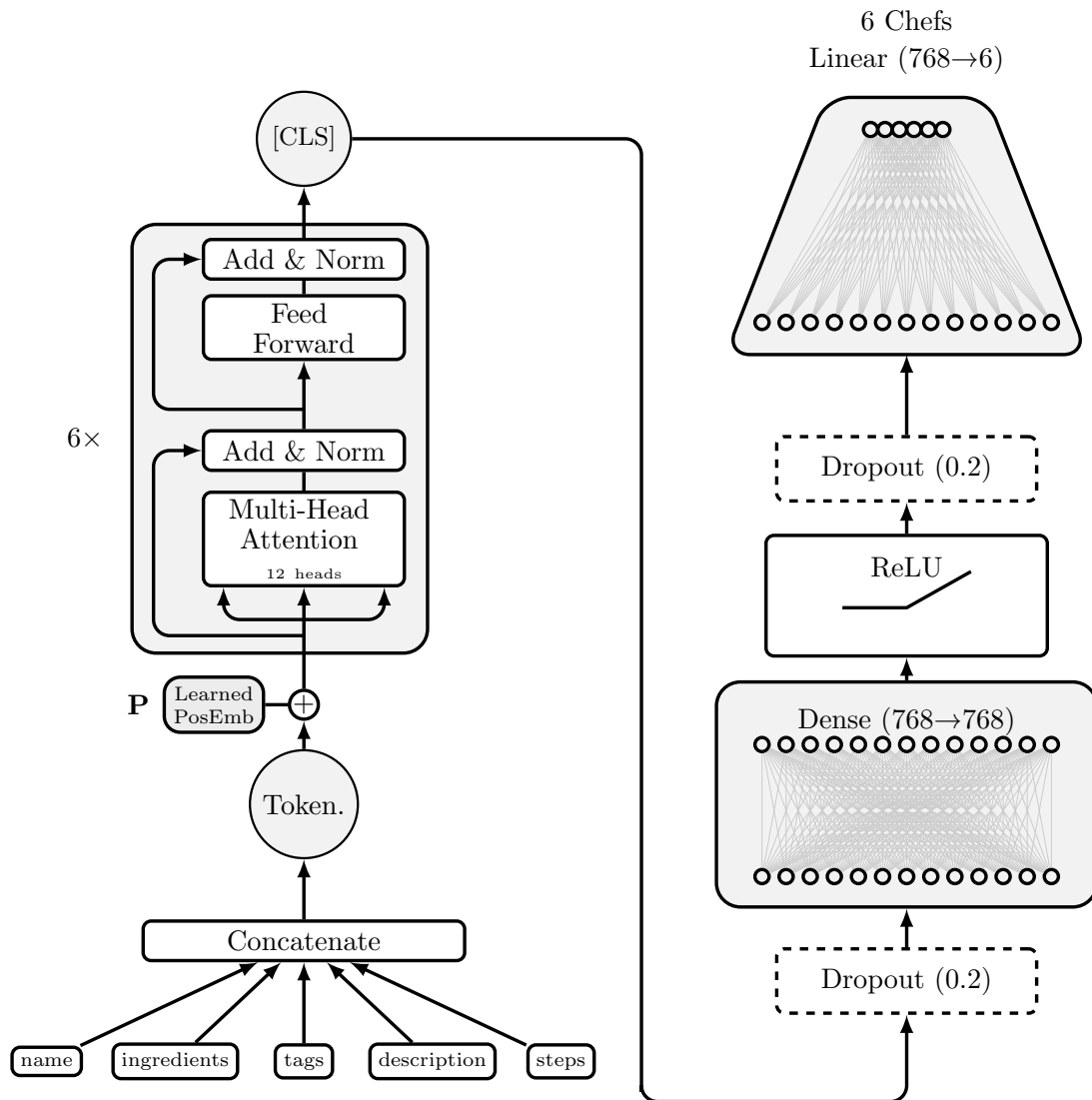


Figure 1: DistilBERT text-only pipeline from concatenated recipe fields to chef prediction.

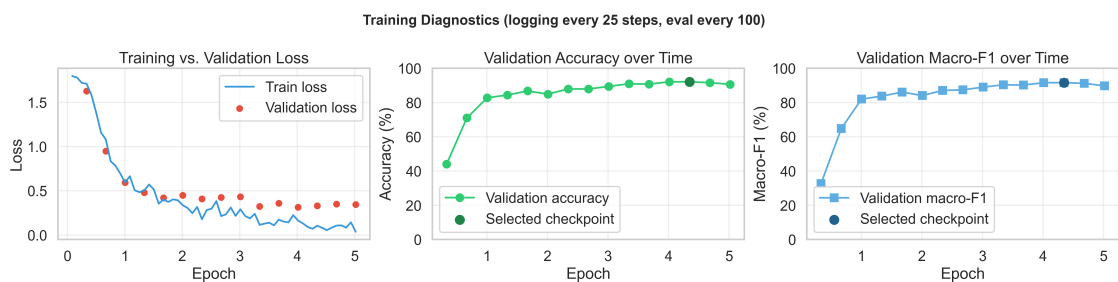


Figure 2: Training loss plus validation accuracy and macro-F1 logged every 25 steps (evaluation every 100).

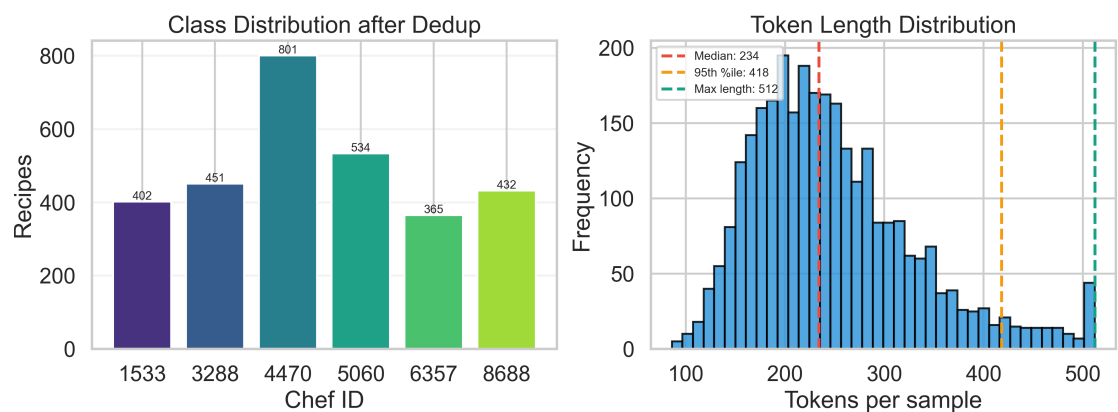


Figure 3: Deduplicated class counts (left) and token length histogram (right).

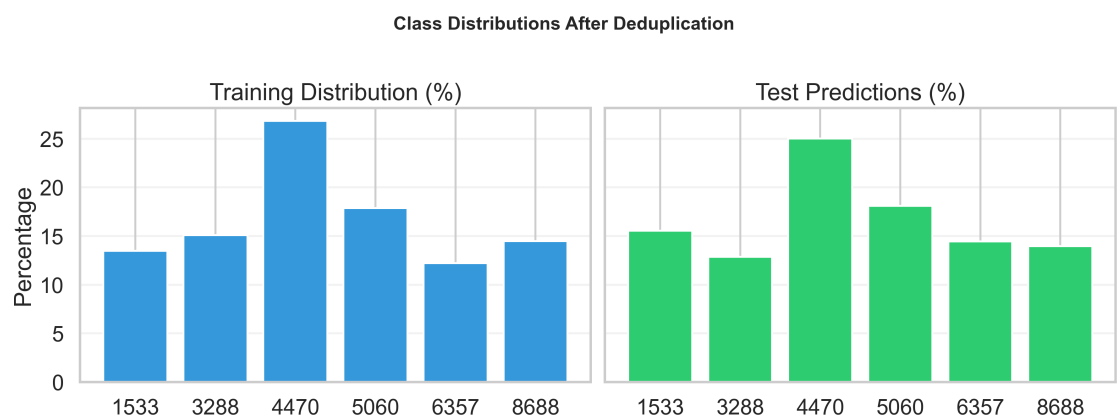


Figure 4: Training distribution (left) vs. blind test predictions (right).

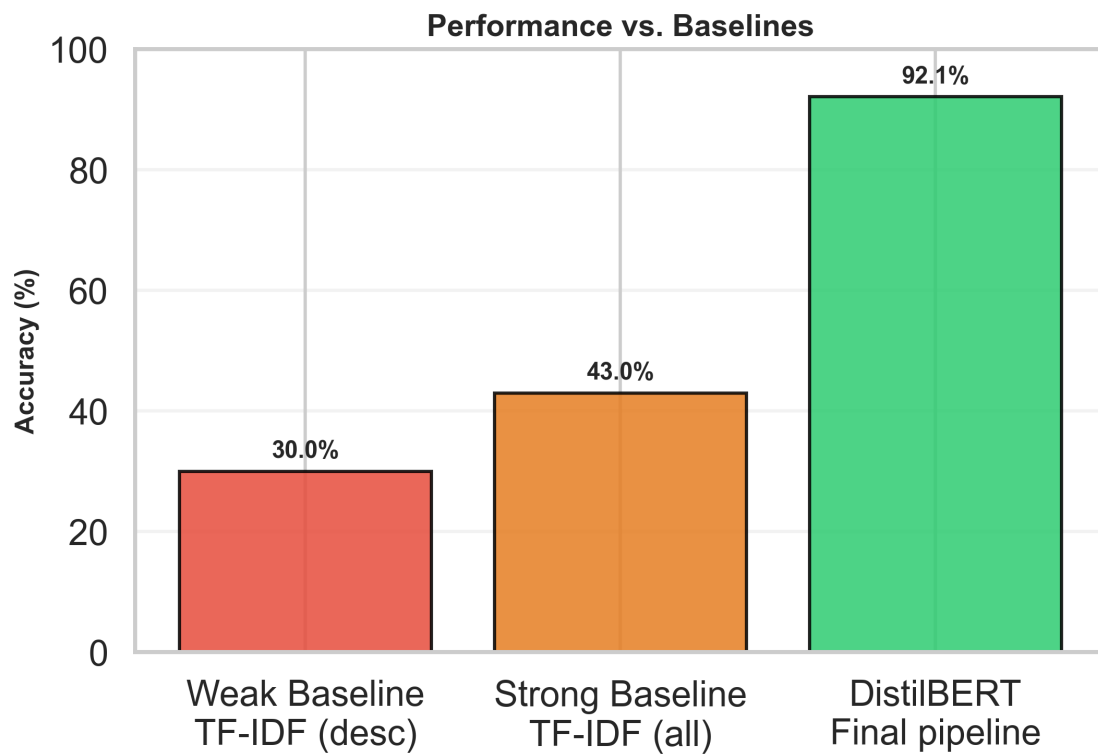


Figure 5: Accuracy gap between TF-IDF baselines and the final DistilBERT pipeline.